Tutorial: Building the OCC Bottle in KodaCAD

This walks through building the classic OCC Bottle (https://occt3d.com/dev/doc/overview/html/occt__tutorial.html) (minus the threads) -- the shape used in OpenCascade's own canonical tutorial -- entirely with KodaCAD's own tools. It's meant to double as onboarding material for a new user and as a regression test: each step is annotated with what it exercises, and the whole sequence is worth re-running after any change that touches sketching, Create/Modify, or Undo/Redo.

What you'll need

A fresh (empty) KodaCAD session. No file import required -- this is built entirely from scratch.

What this exercises

Workplane creation, the AIS ViewCube, the RPN calculator's send-to-KodaCAD workflow, H/V construction lines, sketch geometry (line, arc, construction circle), Create Empty Part, Pull, Fillet (including its edge-ownership resolution and the sample-and-verify fallback for non-analytic circular edges), Shell, workplane visibility surviving a modification, Undo/Redo across shape-replacing operations, and STEP save/reload (file size, geometry round-trip).

Step 1 -- Create a workplane on the XY plane

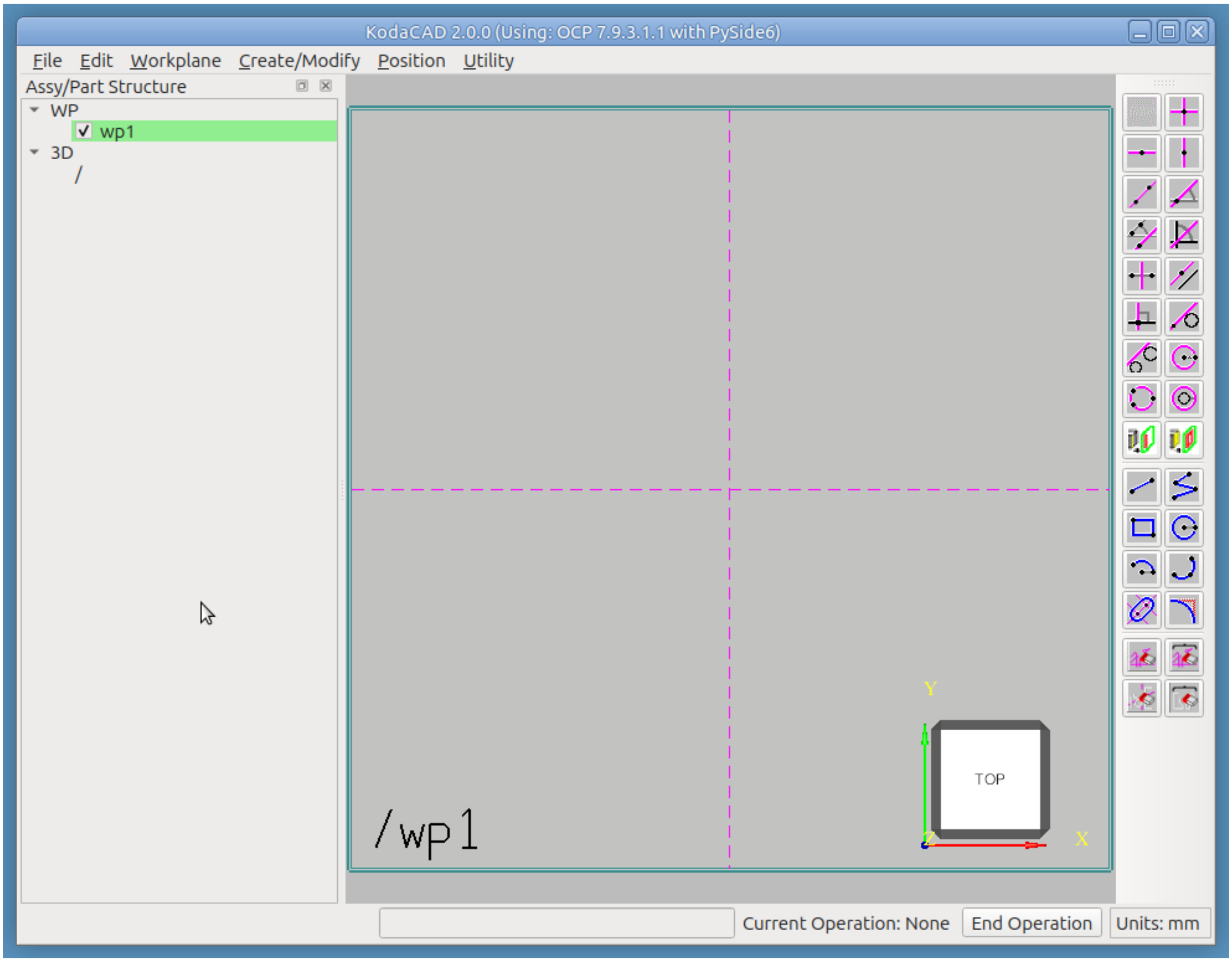
Workplane -> At Origin, XY Plane

- A workplane appears in the X/Y plane, with Horizontal and Vertical construction lines intersecting at the origin, ready to sketch on.

Exercises: workplane creation, the auto-fit border/label cosmetics.

Step 2 -- Orient the view

Click the **top face of the AIS ViewCube** (bottom-right corner of the viewport). The camera animates to a straight-down view of the workplane -- the natural orientation for sketching its profile.



Exercises: ViewCube face-click animation. h-clines

Step 3 -- Lay out reference construction lines with the calculator

This is a fast way to place a set of related H (horizontal) construction lines at computed offsets, using the calculator as a live scratchpad rather than re-typing each value by hand. Hovering the cursor over any tool in the 2D sketch tools panel shows a **tooltip** describing what it does.

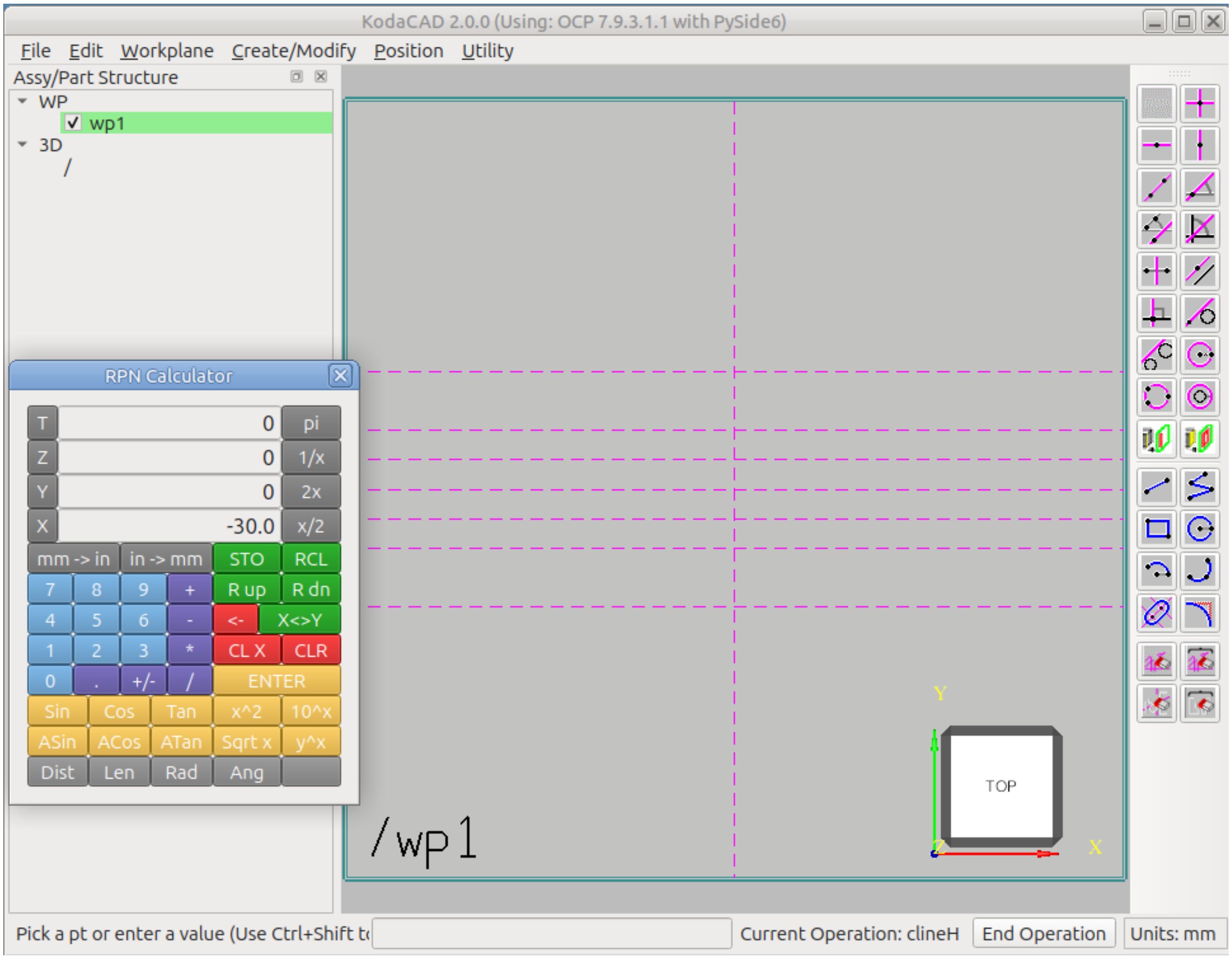
- Click the **Horizontal Construction Line** tool in the 2D sketch tools panel.
  - The status bar prompts *"Pick a pt or enter a value"*.
  - "Current operation: clineH" is confirmed to the right of the status bar.
  - A *proposed* horizontal construction line follows the cursor in the viewport.

- In the **Utility** menu, click **Calculator** to open the calculator.
  - Use the calculator's keys to enter `30`, then click the **X** button (to the left of the value) to send it to KodaCAD -- this forwards the value straight to the armed tool, and an H cline is placed at y = 30.

- Press **x/2** (X register now reads 15), then **X** again -- a second H cline is placed at y = 15.
  - Press **x/2** again (7.5), **X** -- a third cline at y = 7.5.

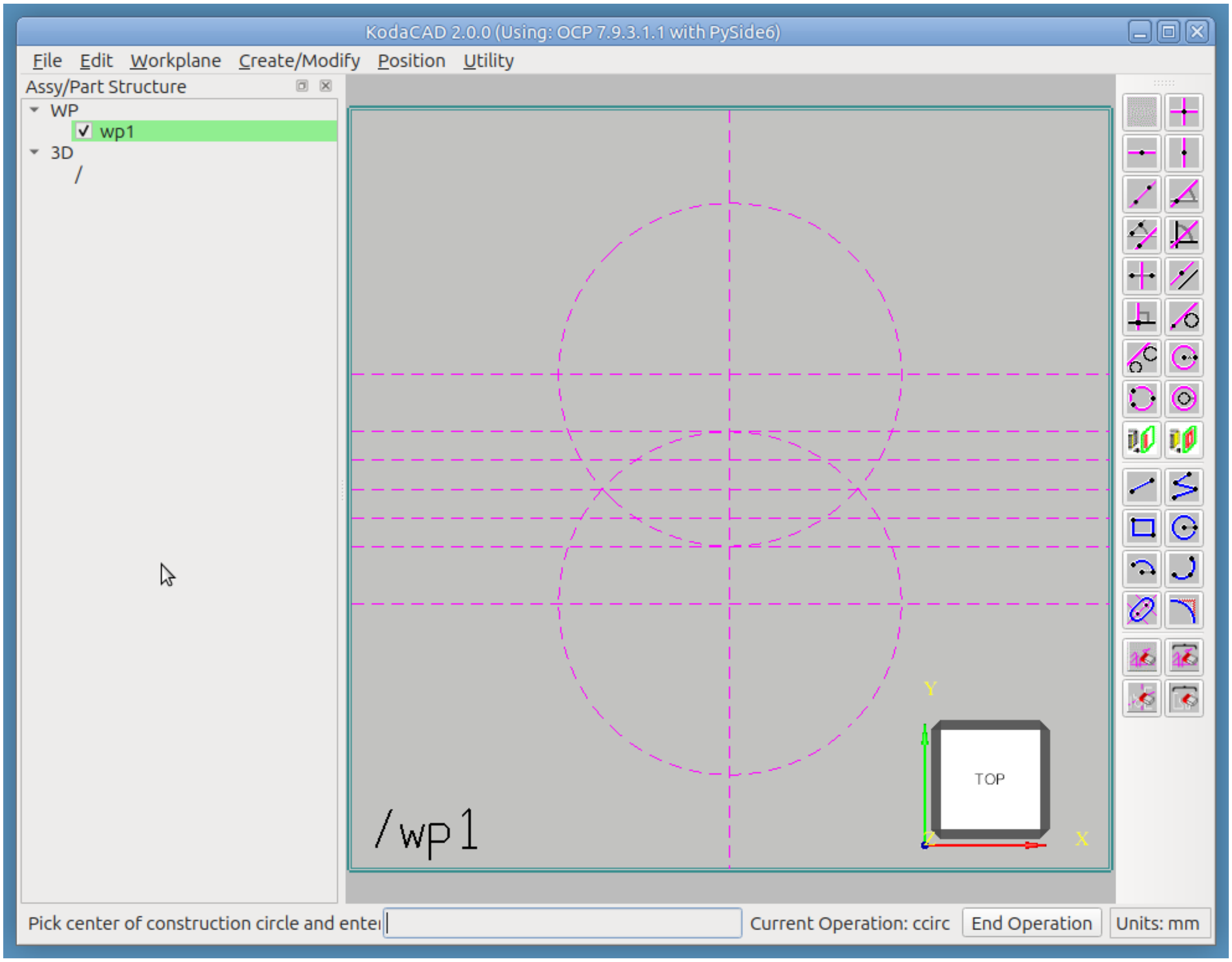
- Press **+/-** (X register flips to -7.5), **X** -- a fourth cline at y = -7.5.
- Press **x2** (X register value doubled), **X** -- a fifth cline at y = -15.

Continue until you have a symmetric set of seven horizontal reference lines at 30, 15, 7.5, 0, -7.5, -15, -30. The H cline tool stays armed across multiple placements without needing to be re-selected. When finished, either middle-click in the viewport or click **End Operation** to exit.



Next, use the **Construction Circle** tool in the 2D sketch tools panel to add two construction circles as shown below. Follow the status bar's guidance: click the center, then a point on the circle. When the cursor hovers near an intersection of construction geometry, a small yellow square glyph appears -- those are the only places a click is accepted. This supports one of KodaCAD's basic paradigms:

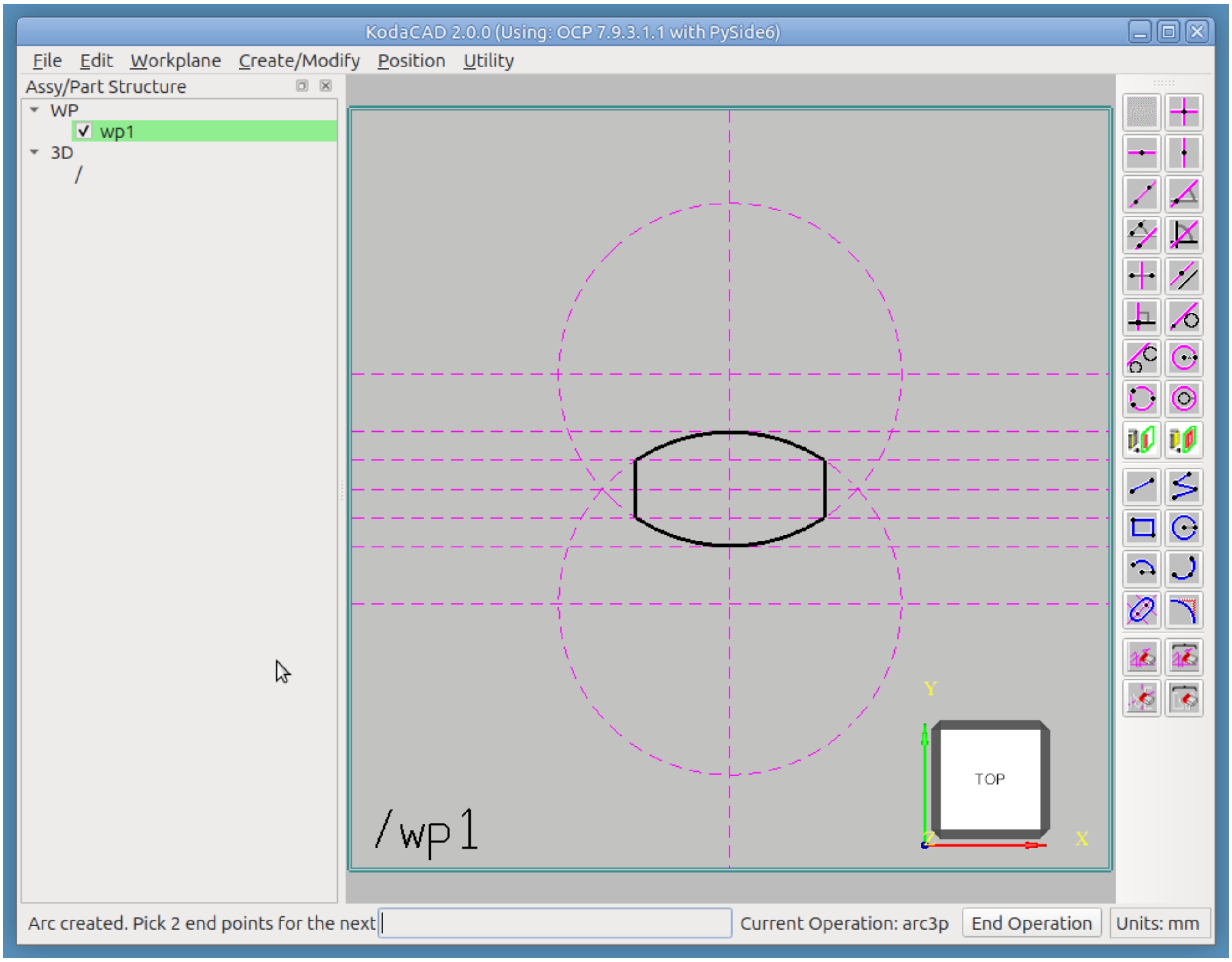
> In KodaCAD, accuracy is built into the sketch. You don't just make a freeform sketch and then add dimensions and constraints later.



Exercises: the calculator's send-to-KodaCAD mechanism (the T/Z/Y/X buttons -> valueFromCalc -> the currently-armed tool's callback), x/2, x2, +/-, and chained tool use (a sketch tool stays armed across multiple placements without needing to be re-selected); the Construction Circle tool; the snap-engine catch glyph (the yellow square) and its role in constraining picks to valid catch points only.

Step 4 -- Sketch the profile

Use the **Line** tool in the 2D sketch tools panel to create two vertical geometry-type line segments, as shown below. After drawing these two lines, clicking the **Arc by 3 Points** tool automatically ends the Line operation and begins the new one, ready to sketch the two arcs. Again, follow the status bar's guidance -- click the arc's end points, then a point on the arc. These four elements form a **closed profile** of geometry-type lines, which will be pulled into the 3D shape in the next step.



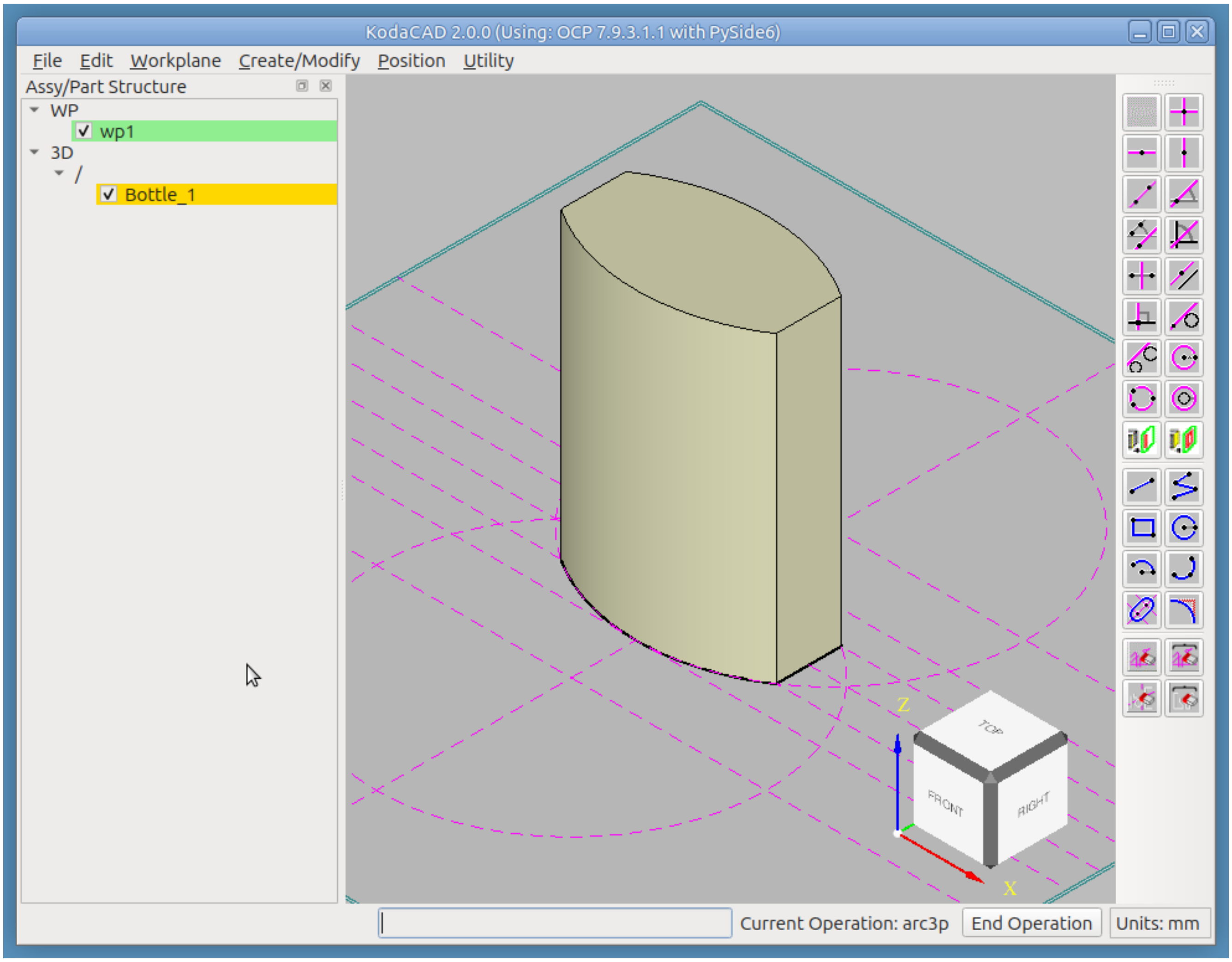
Exercises: Line and Arc sketch tools; snap-to-construction-line catching.

Step 5 -- Create the Bottle from the profile

- RMB click on the '/' and select **Create Empty Part**, name it 'Bottle'
  - The empty part is created in the tree as a child of '/'
  - It is highlighted in yellow, signifying that it has ben set **Active**

- **Create/Modify -> Pull**. The Pull dialog is displayed. Set the parameters as follows:
  - Operation: Add Material
  - Mode: Linear
  - Distance (mm): 70

- Click Done
- Clicking a corner of the ViewCube in the viewport switches to an isometric view.



Exercises: Pulling a profile linearly into a solid.

Step 6 -- Fillet the edges

The first workplane is no longer needed at this point. **Uncheck its box in the tree to hide it -- don't delete it yet.** (A second workplane is coming up in Step 7, so leaving this one hidden rather than deleted sets up a check worth doing once both workplanes exist: confirm this one is *still* unchecked, not silently re-shown.)

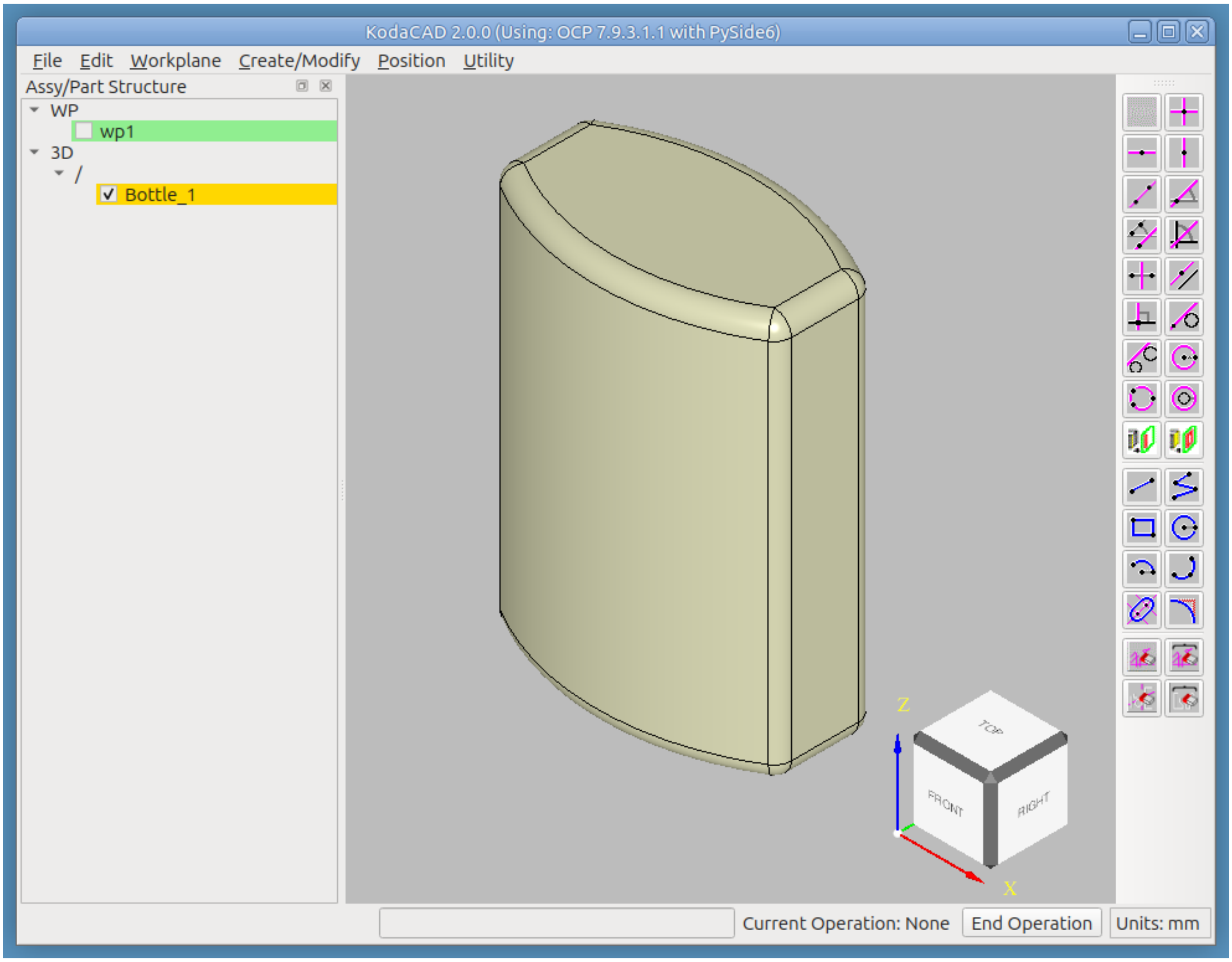
A few ways to adjust the viewport at any point:

- LMB drag to rotate
- MMB drag to pan
- RMB in the viewport -> **Draw -> Fit** to zoom to fill the viewport
- Click a face, edge, or corner of the ViewCube (only works when no operation is currently active)

Then:

- Set the new part active by RMB clicking it in the tree.
- **Create/Modify -> Fillet**.
- Select all 12 edges of the bottle, one at a time -- the status bar acknowledges each one.

- Enter `3` as the radius.
- **Check the tree: confirm the first workplane is still hidden.** Any modification rebuilds the tree, which is exactly the moment a hidden workplane could silently reappear if its visibility state weren't actually being respected.



Exercises: fillet's edge-ownership resolution at pick time; the sample-and-verify circumcenter fallback if any of these edges aren't typed as an analytic circle/line after display prep; workplane visibility surviving a modification -- a hidden workplane used to silently re-check itself on every tree rebuild, which every modification triggers.

Step 7 -- Create a workplane on the bottle's top face

- **Workplane -> On Face**.
- Click the top face of the bottle to set the UV plane.
- Click one of the flat side faces to set the +U direction.
- A small workplane appears on the bottle's top face, its origin at the face's geometric center.

Exercises: workplane-on-face.

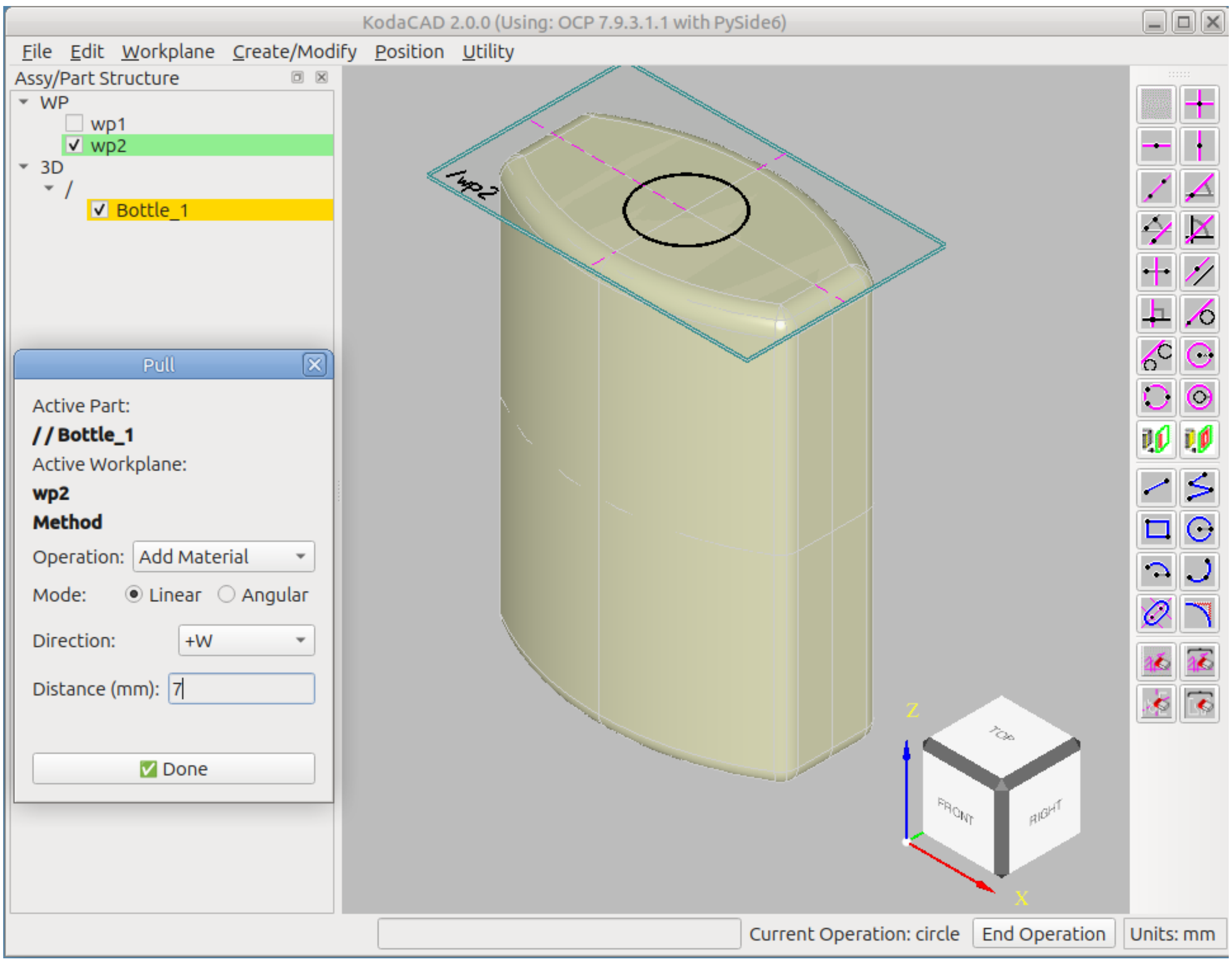
Step 8 -- Create the neck of the bottle

Sketch the circle profile, centered on the top face:

- Use the **Circle** tool.
- Click the intersection of the H and V construction lines.
- Enter `7.5` as the radius.

Pull the profile upward, with the bottle still active (highlighted in the tree):

- **Create/Modify -> Pull**.
- In the dialog: choose **Add Material**, **Linear**, direction **+W**, Distance **7**
- Click Done

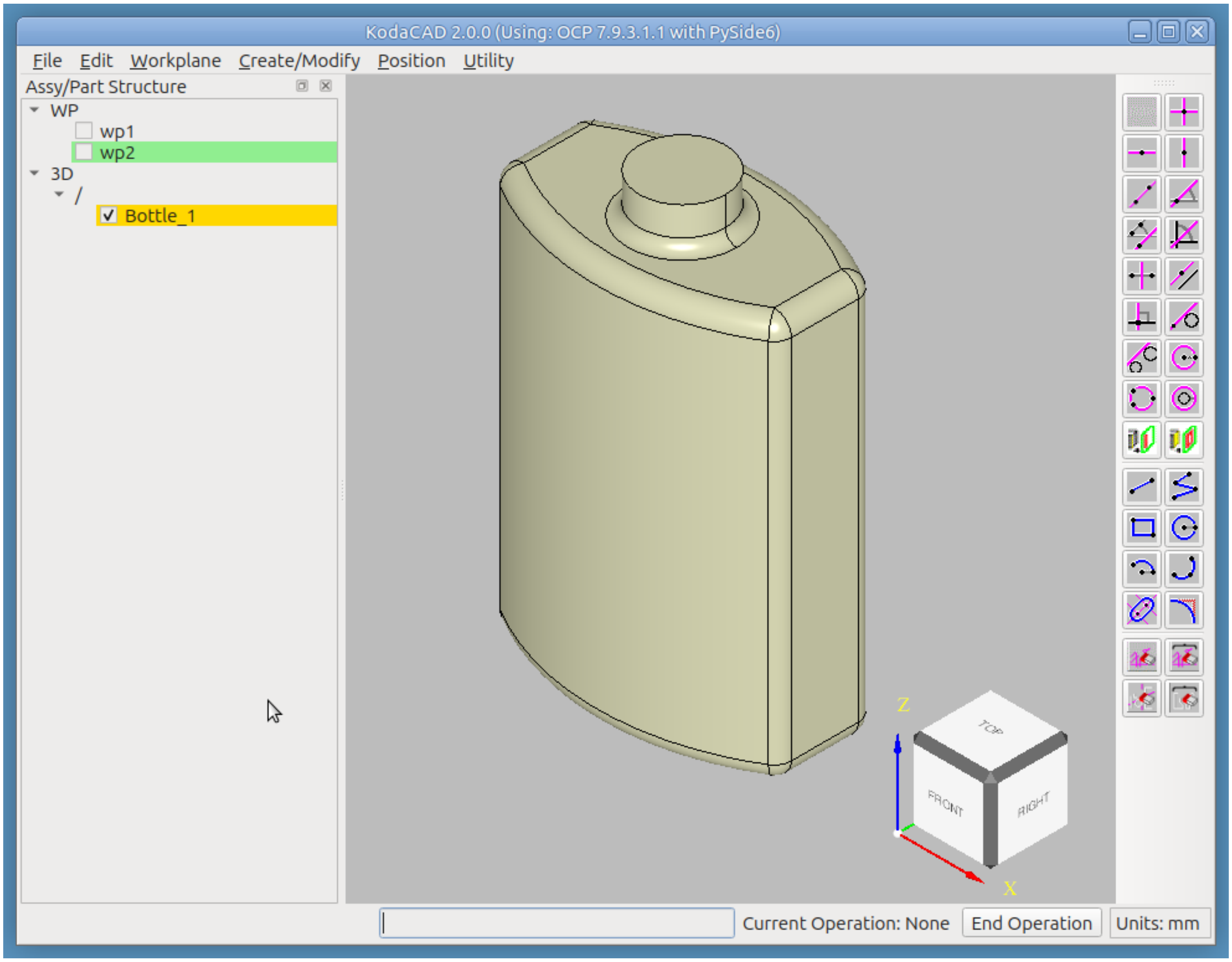


Exercises: workplane-based circle sketching feeding directly into Mill/Pull; a second, feature-adding operation on an already-filleted part.

Step 9 -- Fillet the neck / top face

Hide this second workplane too (uncheck its box) -- it's no longer needed. **Confirm both workplanes now**

show as unchecked in the tree, and stay that way through the fillet below. With the bottle still active, add a 2 mm radius fillet at the base of the neck.

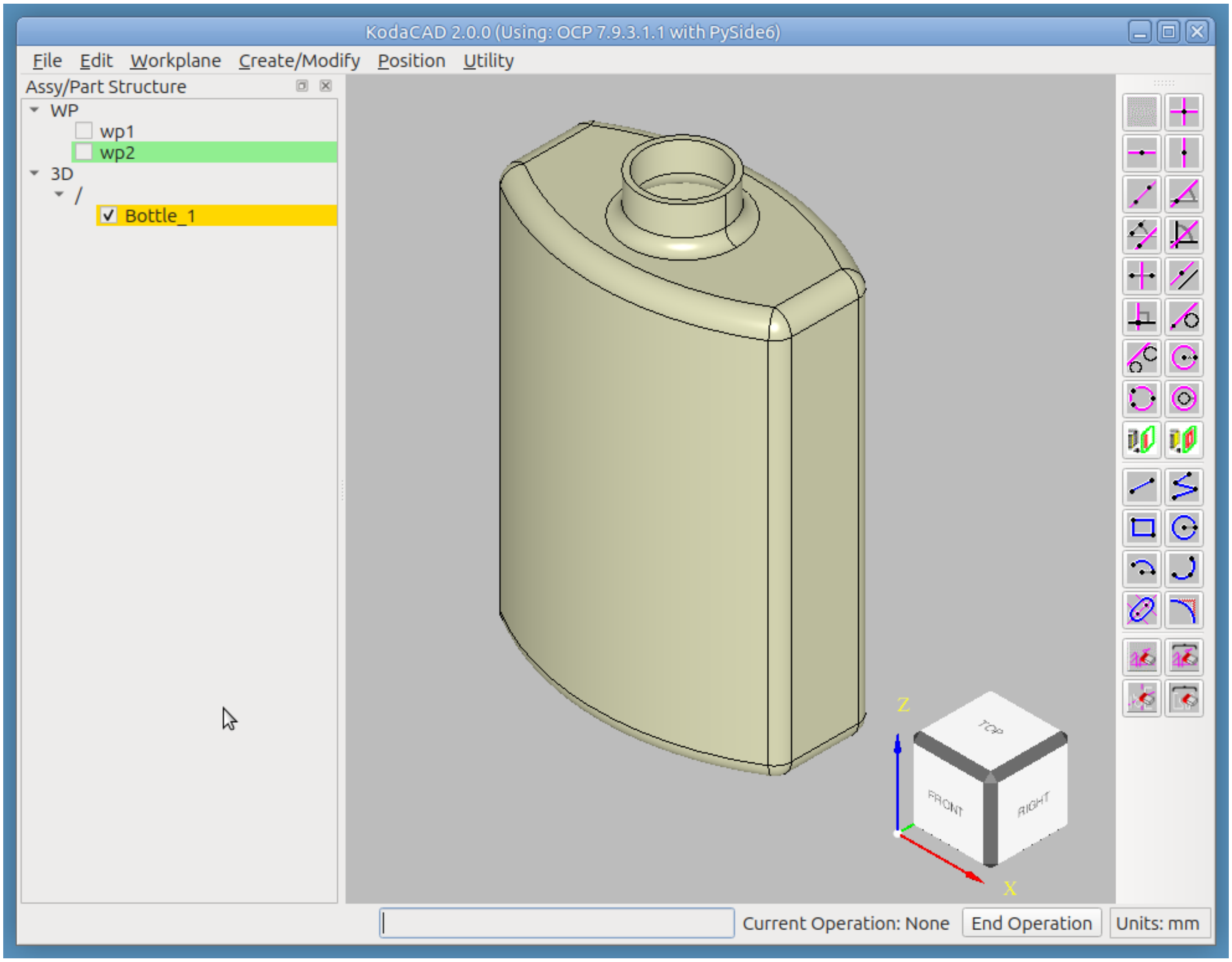


Exercises: a second fillet operation on the same part, at a different location and radius than Step 6's; workplane visibility surviving a modification, now with two hidden workplanes at once rather than one.

Step 10 -- Shell the body

The final step is to shell the bottle, leaving the top open. With the bottle still active:

- **Create/Modify -> Shell**.
- Select the top face of the neck (the opening the bottle should keep).
- Enter `1` as the wall thickness.



Exercises: face selection for Shell, `replace_shape` *'s propagation to the display.*

Step 11 -- Undo the last few steps

Undo the Shell, then the neck fillet, then the Pull, then the first fillet, one at a time, confirming each one visibly reverts before redoing back to the finished bottle.

Exercises: undo/redo's shape-vs-location redraw distinction -- Shell, Fillet, and Pull are all shape-replacing operations that don't move the part, the specific case that needed its own fix (recording the touched prototype's entry) separately from the location-comparison fast path.

Step 12 -- Save and reload

Save the session as a STEP file, then reload it (or open it in a second viewer) and confirm the geometry, and file size, both look right.

Exercises: the STEP writer's PCURVE-suppression setting; general save/reload fidelity.

Notes for whoever runs this next

- If any step's status bar message, menu label, or tool name has changed since this was written, that's worth fixing here directly -- this document drifting out of sync with the actual UI is exactly the kind of thing it's meant to catch.
- Consider adding a `Rad / Ang` measurement step once the bottle is built -- e.g. measure the neck's radius and confirm it matches what was sketched, exercising the calculator's measurement tools against a freshly-created (not STEP-imported) part.
- See also the Assembly Structure tutorial (../as1-oc-214/README.md) and the Quaoar Chassis tutorial (../chassis/README.md) for KodaCAD's assembly machinery -- this tutorial is deliberately single-part, and doesn't touch shared instances, reparenting, or the Position dialog at all.
- See the Jack-o'-Lantern tutorial (../jack/README.md) for a fuller workout of the 2D sketch toolbar, and for this same pull-then-heavily-fillet technique reused to build a pumpkin, followed by a real, multi-profile Pull operation this tutorial's own single-profile neck-pull doesn't demonstrate.